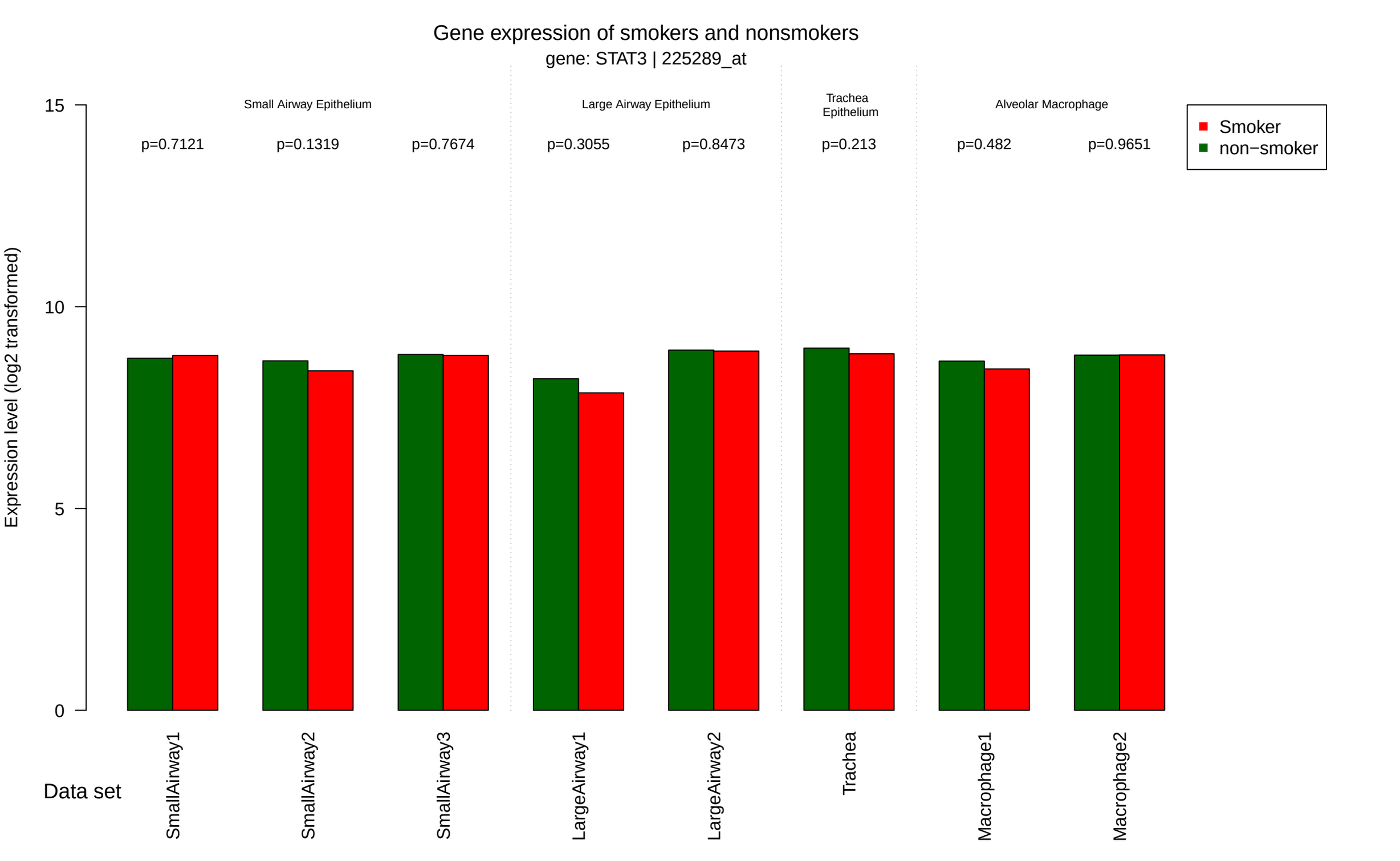
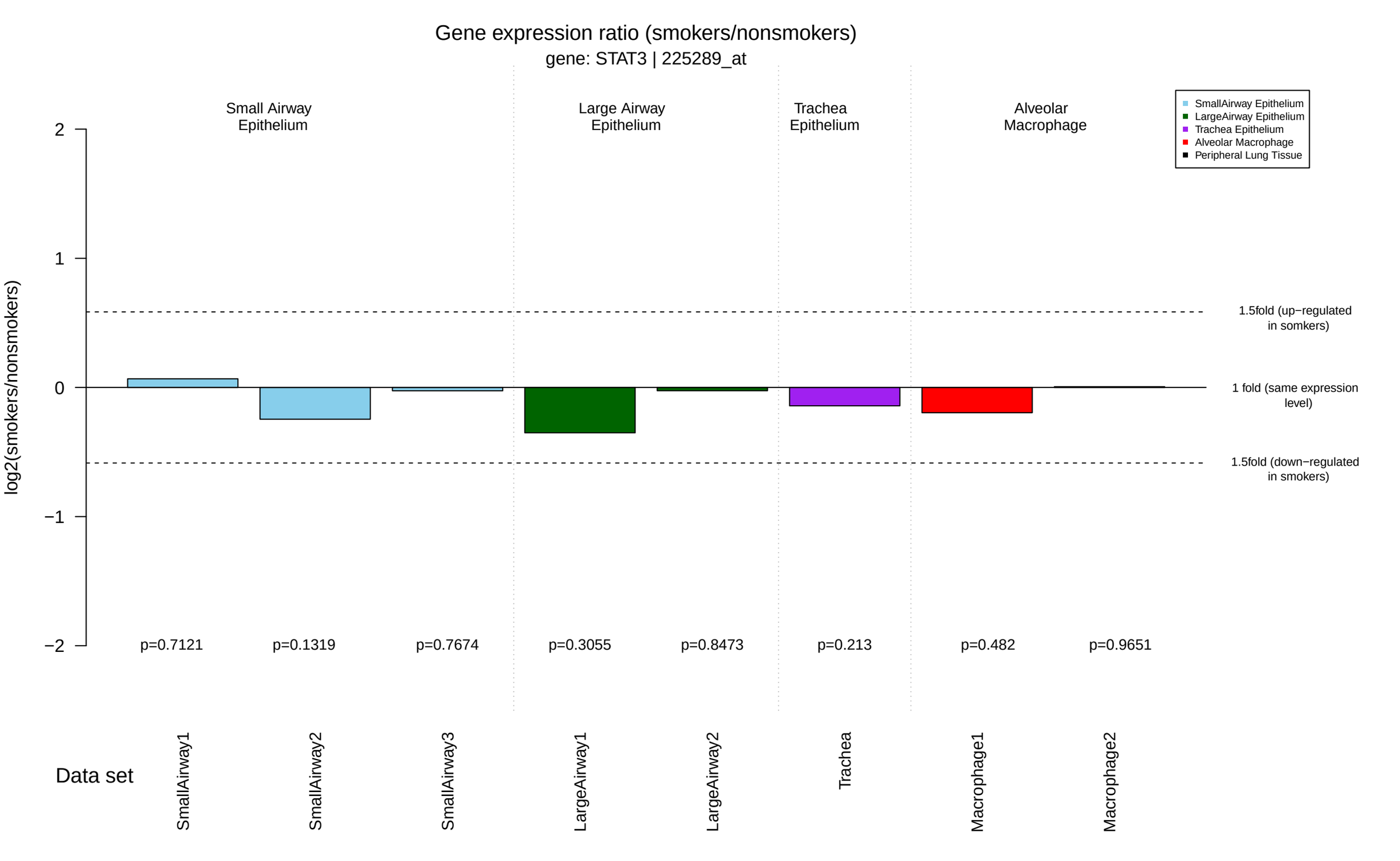
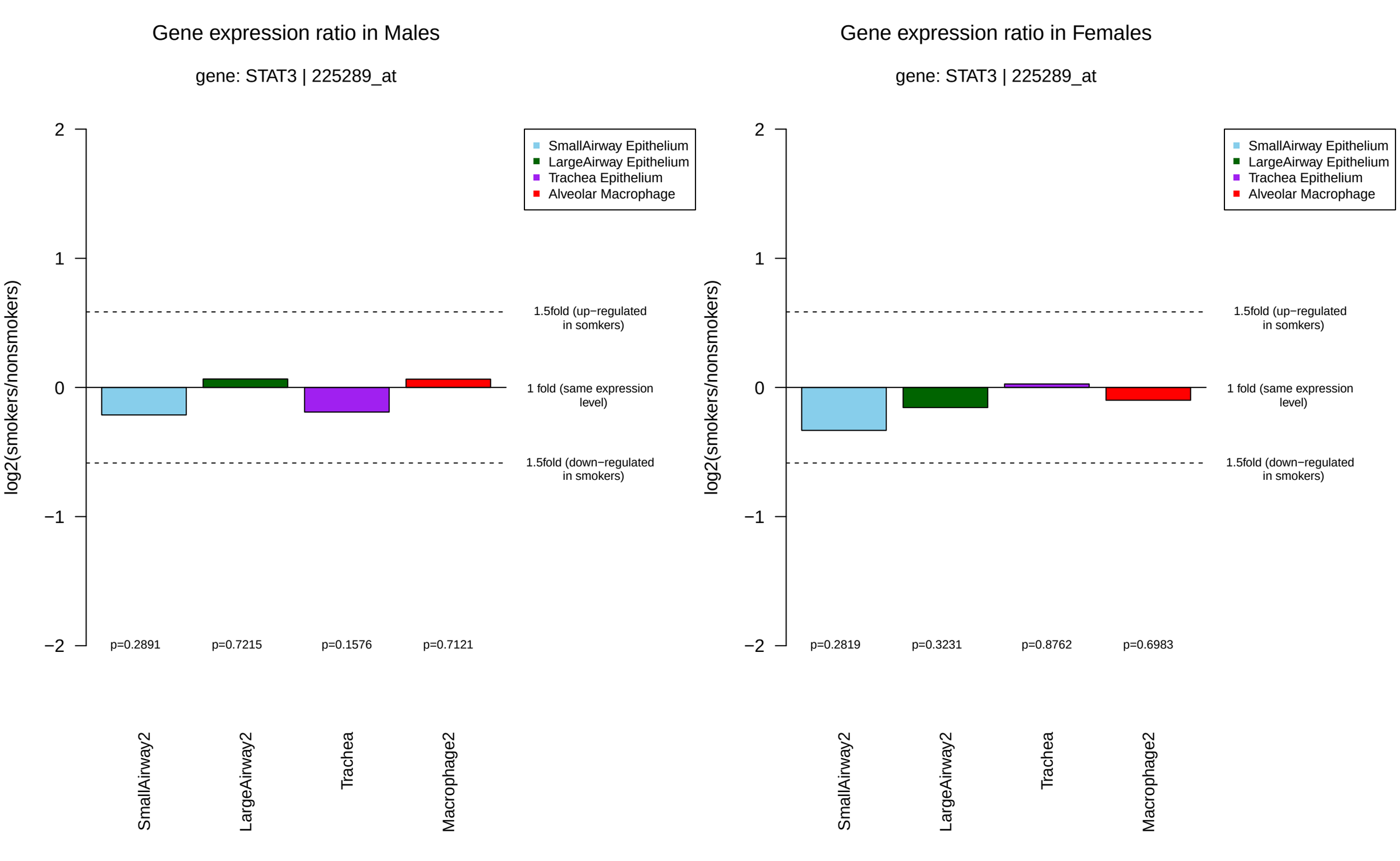
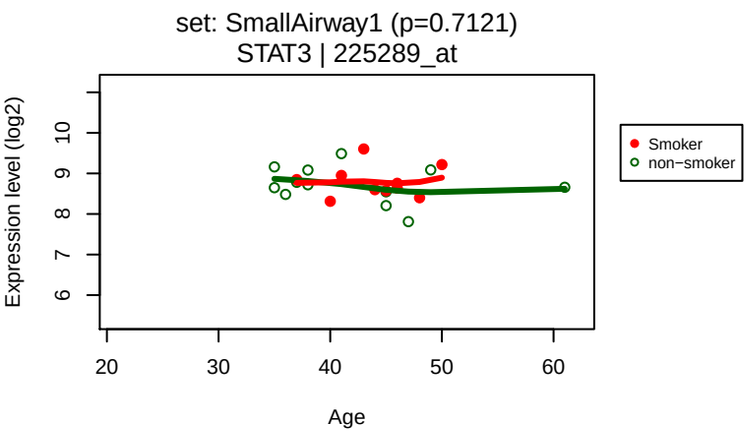
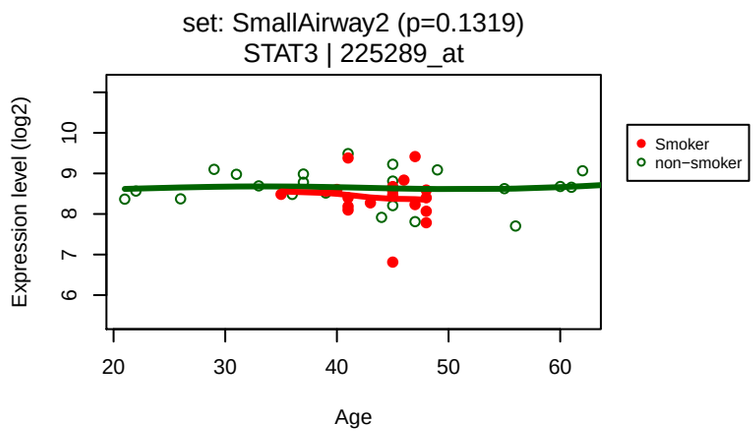
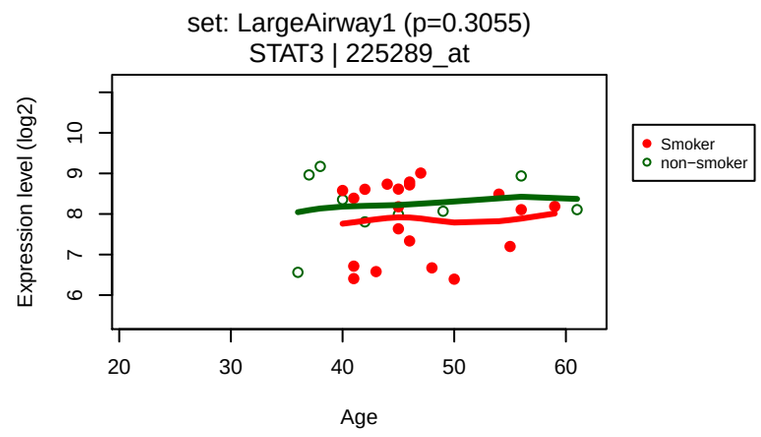

Gene expression of smokers and nonsmokers
gene: STAT3 | 225289_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7121
p=0.1319
p=0.7674
p=0.3055
p=0.8473
p=0.213
p=0.482
p=0.9651
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

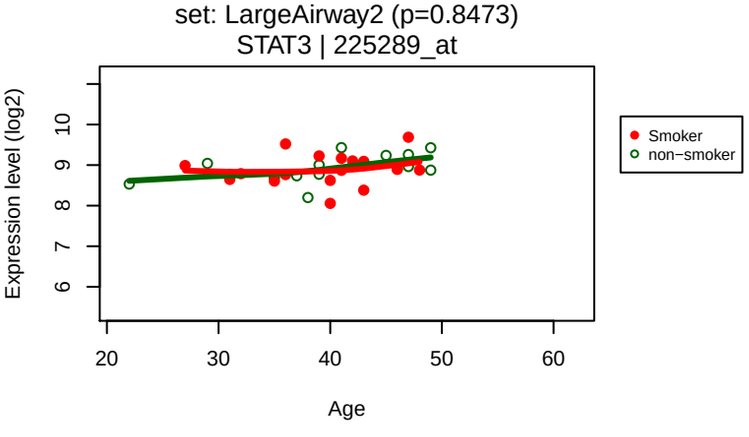

Gene expression ratio (smokers/nonsmokers)
gene: STAT3 | 225289_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7121
p=0.1319
p=0.7674
p=0.3055
p=0.8473
p=0.213
p=0.482
p=0.9651
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

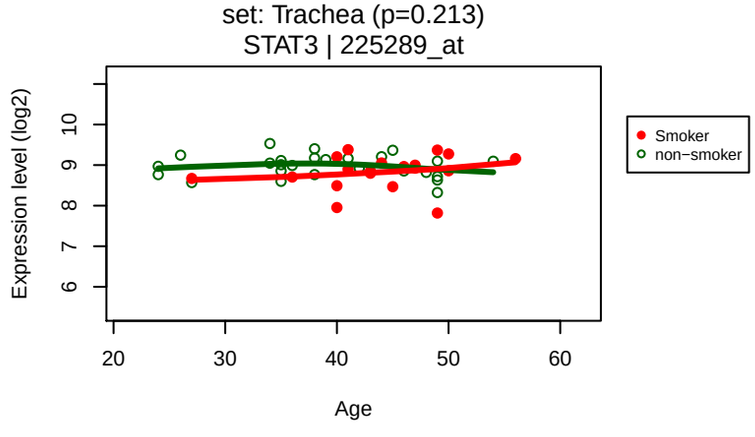

Gene expression ratio in Males
gene: STAT3 | 225289_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2891
p=0.7215
p=0.1576
p=0.7121
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAT3 | 225289_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2819
p=0.3231
p=0.8762
p=0.6983
SmallAirway2
LargeAirway2
Trachea
Macrophage2

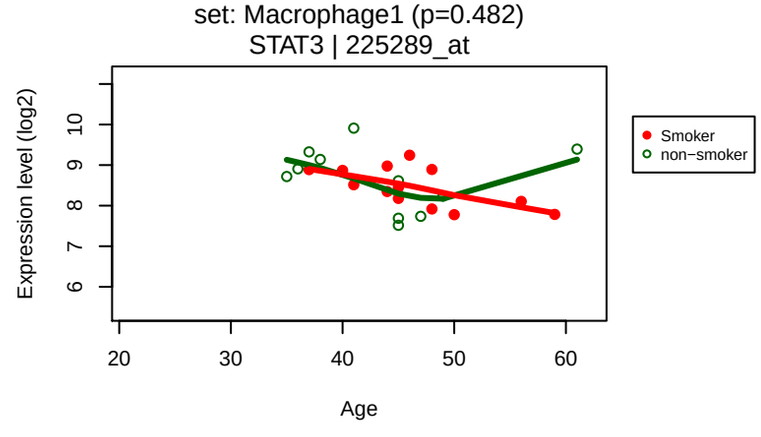

set: SmallAirway1 (p=0.7121)
STAT3 | 225289_at
Expression level (log2)
Age
Smoker
non−smoker

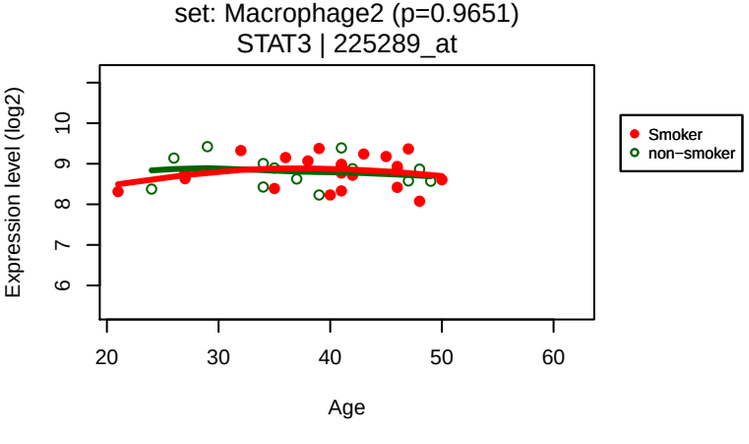

set: SmallAirway2 (p=0.1319)
STAT3 | 225289_at
Expression level (log2)
Age
Smoker
non−smoker

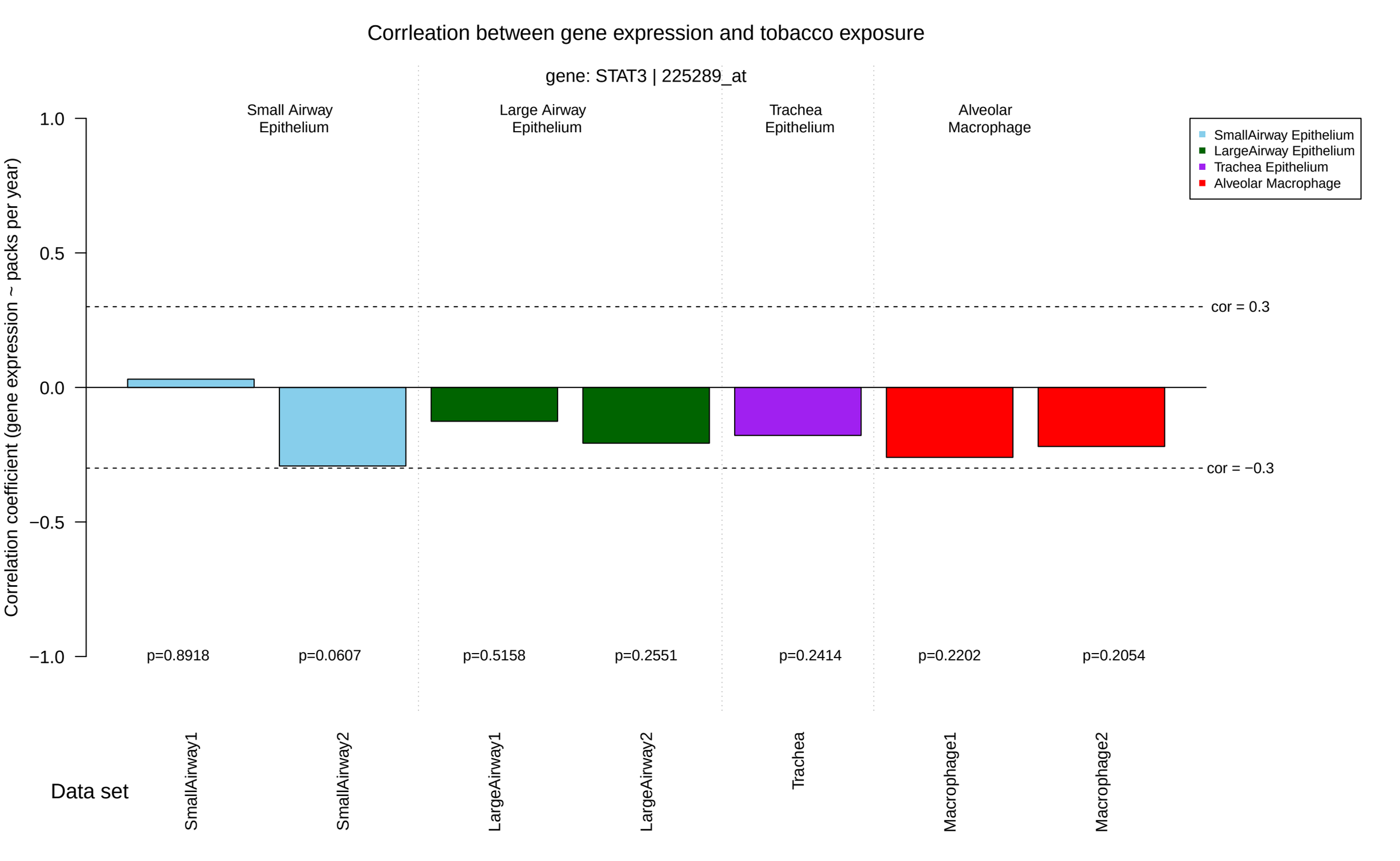

set: LargeAirway1 (p=0.3055)
STAT3 | 225289_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8473)
STAT3 | 225289_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.213)
STAT3 | 225289_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.482)
STAT3 | 225289_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9651)
STAT3 | 225289_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAT3 | 225289_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8918
p=0.0607
p=0.5158
p=0.2551
p=0.2414
p=0.2202
p=0.2054
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2